

# Product Specification Document

**Project:** News Aggregator System

**Version:** 1.0

**Prepared By:** Development Team

## 1. Purpose

The **News Aggregator System** is a web-based application designed to gather, summarize, and present the latest news from multiple sources in real time. The system's primary goal is to deliver credible, concise, and categorized news to users through an interactive interface. It minimizes misinformation by leveraging AI-based summarization and credibility assessment modules.

## 2. Scope

This product provides users with an intuitive way to access the latest news, filter by category, and explore summarized content. The system fetches articles from public APIs, analyzes the data, and displays it within a responsive user interface. The News Aggregator System supports integration with automated testing platforms for quality assurance, making it ideal for deployment on web environments.

## 3. Product Overview

The News Aggregator System is developed using **Vite, React, TypeScript**, and **TailwindCSS**. It connects to external APIs to fetch real-time news, processes the data through a summarization and fact-checking module, and displays it via a dynamic frontend interface. The application is lightweight, fast, and optimized for both desktop and mobile browsers.

## 4. Key Features

| Feature                  | Description                                                            |
|--------------------------|------------------------------------------------------------------------|
| News Aggregation         | Collects real-time news data from multiple verified APIs.              |
| AI Summarization         | Condenses lengthy articles into short summaries for readability.       |
| Category Filtering       | Allows filtering news by category such as Politics, Tech, Sports, etc. |
| Search Functionality     | Provides instant search for topics and keywords.                       |
| Credibility Score        | Displays a reliability index based on news source and metadata.        |
| Responsive UI            | Ensures seamless performance on both mobile and desktop devices.       |
| Dark Mode                | Offers theme customization using Tailwind CSS configurations.          |
| Performance Optimization | Uses Vite for faster build and optimized load times.                   |

## 5. System Architecture

The system follows a client-centric architecture where the frontend communicates directly with external APIs. The core modules include:

- Frontend UI Layer (React + TailwindCSS)

- Data Aggregation Layer (API Fetch Service)
- Summarization & Credibility Analysis
- Presentation Layer (Card-based news display)
- Build & Deployment Layer (Vite + Bun)

## 6. Functional Requirements

1. The homepage must load a list of trending news articles. 2. Each news card must include the title, image, source, and timestamp. 3. Articles should display summarized content on selection. 4. The user can search and filter articles dynamically. 5. The system should fetch updates from APIs every 10–15 minutes. 6. The application must maintain responsiveness across screen sizes.

## 7. Non-Functional Requirements

- Performance: Average page load < 3 seconds
- Scalability: Supports addition of new APIs without redesign
- Security: API keys stored securely in environment variables
- Usability: Minimal scrolling, intuitive card layout
- Maintainability: Modular codebase using TypeScript
- Testing: Compatible with automated front-end testing frameworks

## 8. System Workflow

1. API Request → Fetch news data from sources using API key. 2. Data Processing → Filter and categorize news. 3. Summarization → Generate concise summaries. 4. Rendering → Display on user interface using reusable components. 5. User Interaction → Enable search, category selection, and detailed views.

## 9. Testing Scenarios

| Test ID  | Description                   | Expected Result                   |
|----------|-------------------------------|-----------------------------------|
| UI-001   | Check homepage rendering      | Homepage loads without API errors |
| UI-002   | Verify dark/light mode toggle | Theme changes correctly           |
| API-001  | Validate API response         | 200 OK and valid JSON             |
| FUNC-001 | Check search functionality    | Search returns matching results   |
| PERF-001 | Measure page load speed       | < 3 seconds                       |

## 10. Future Enhancements

- Integrate AI-based sentiment analysis for tone classification.
- Add user authentication for personalized feeds.
- Introduce push notifications for breaking news.
- Enable offline reading mode using caching.
- Develop mobile app version using React Native.

## 11. Conclusion

The **News Aggregator System** efficiently collects, processes, and presents news content in real time. It combines modern web technologies, responsive design, and intelligent summarization to deliver credible and accessible news. The architecture ensures easy integration with future AI modules, API expansions, and automated testing frameworks.